Module Fan v1.1

SKU:M013-V11

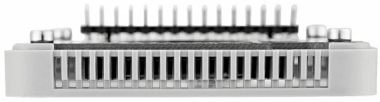




















Description

Module Fan v1.1 is a cooling module designed for the M5Stack stacking system. It features an STM32 microcontroller and uses the **I2C** protocol for control. It supports **reading and controlling** the fan speed and includes a **stall protection** function. This module is primarily used for cooling in stacking systems, effectively improving the system's thermal performance. It is suitable for embedded devices, smart hardware, and other applications requiring thermal management.

Features

- M5Stack stacking system
- STM32 microcontroller
- I2C protocol control
- Fan speed reading and control
- Stall protection function
- Development platforms
  - UiFlow1
  - UiFlow2
  - Arduino IDE

Includes

- 1 x Module Fan v1.1

Applications

- Cooling for M5Stack devices

Specifications

Specification	Parameters
MCU	STM32F030F4P6
Communication	I2C(0x18)
Operating Voltage	5V
Noise Test (Ambient Noise 40.8dB)	PWM20%:48.6dB PWM60%:64.5dB PWM100%:70.6dB
Operating Power	PWM20%:DC 5V@13.52mA PWM60%:DC 5V@56.19mA PWM100%:DC 5V@125.14mA
Standby Power	DC 5V@503.49uA
Fan Speed	11500RPM±10%
Maximum Airflow	1.23 CFM
Fan Speed Control	Supports reading fan speed and PWM speed control
Stall Protection	Supports stall protection function
Operating Temperature	0~40°C
Product Size	54.0 x 54.0 x 13.1mm
Product Weight	14.7g
Package Size	95 x 184 x 15.7mm
Gross Weight	28.3g

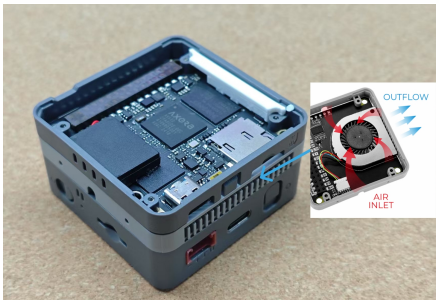
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Stall Protection Note

If the fan is stalled due to foreign objects, it will automatically resume operation after the obstruction is removed. However, prolonged stalling may damage the fan or even cause it to burn out. Therefore, please promptly clear any blockages and ensure the fan operates normally.

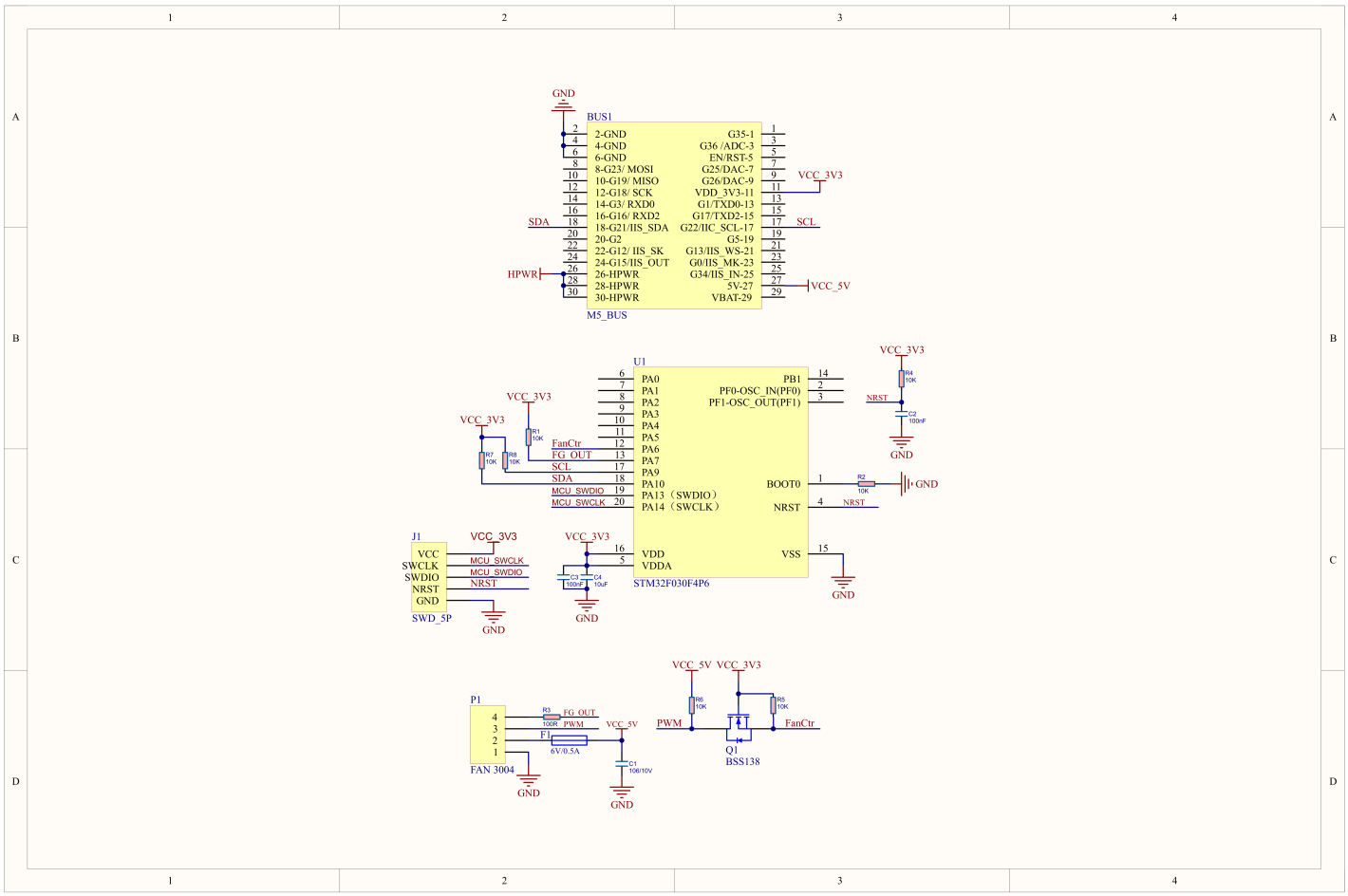
Cooling Module Installation

The air intake of Module Fan v1.1 is located at the bottom of the module. For optimal cooling performance, it is recommended to stack it on top of the module that requires cooling, as shown in the image below.



Schematics

- Module Fan v1.1 Schematics PDF



PinMap

Fan

Fan	VCC	GND	FG_OUT	PWM
Color	Red	Black	Yellow	Blue

> FG_OUT is the fan speed waveform output pin, and PWM is the fan speed control pin.

STM32

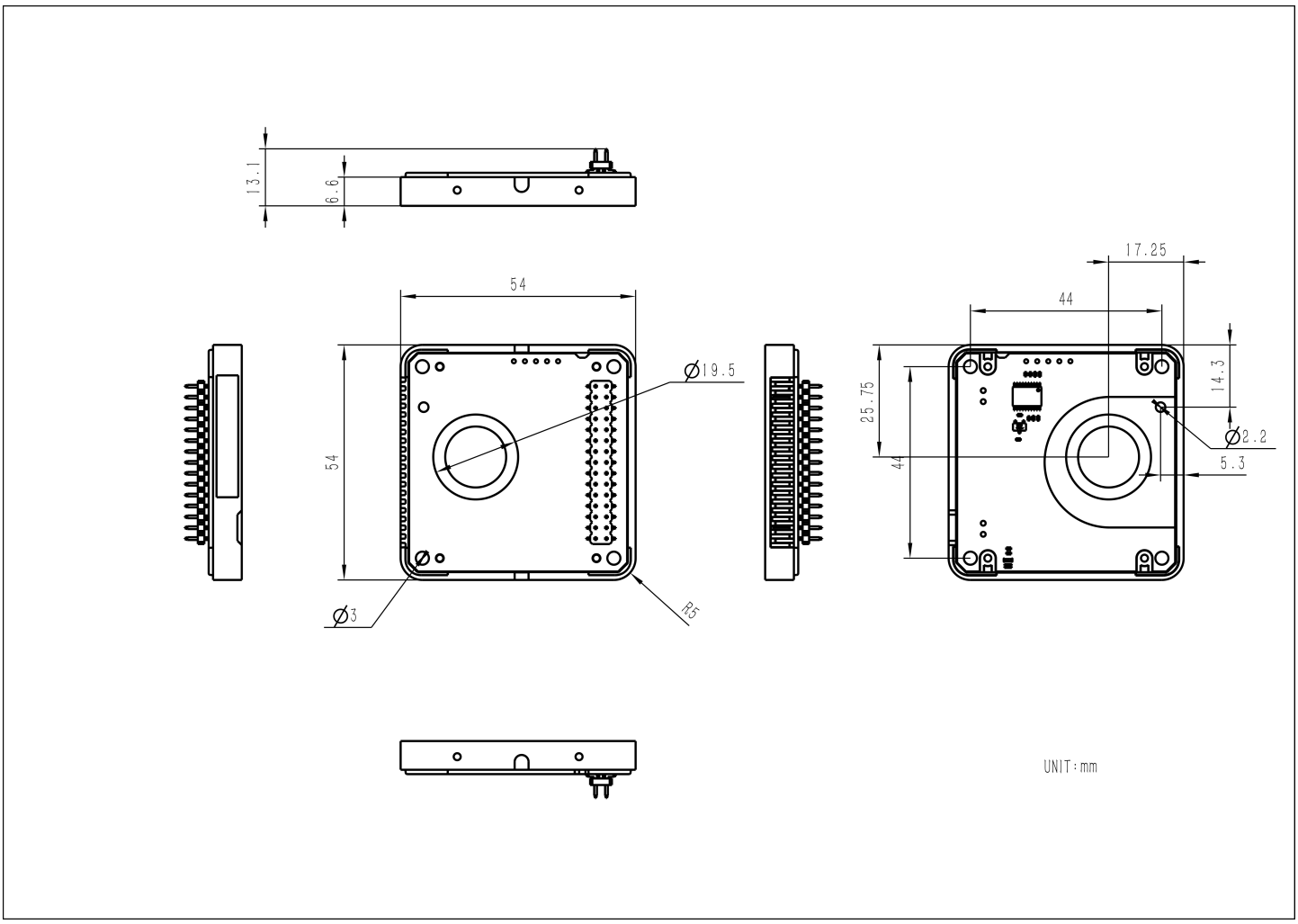
STM32F030F4P6	PA6	PA7	PA9	PA10
Fan	FanCtr	FG_OUT	SCL	SDA

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
	7	8	
	9	10	
	11	12	3V3
	13	14	
	15	16	
SDA	17	18	SCL
	19	20	
	21	22	
	23	24	
HPWR	25	26	
HPWR	27	28	5V
HPWR	29	30	

Model Size

- Module Fan v1.1 Model Dimensions PDF



Datasheets

- Fan Datasheet

Softwares

Arduino

- Module Fan v1.1 Arduino Guide
- Module Fan v1.1 Arduino Library

UiFlow1

- Coming soon...

UiFlow2

- Module Fan v1.1 UiFlow2 Docs

Internal Firmware

- Module Fan v1.1 Internal Firmware

Protocol

- Module Fan v1.1 I2C Protocol PDF

M5Stack Module-FAN I2C Protocol																		V1 (FW Version) 2024/11/28
REG MAP (Addr: 0x18)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Fan control	0x00 R/W	Control status																Control status: 风扇工作状态 0: 不工作 1: 工作 default: 1
PWM frequency	0x10 R/W	Frequency																Frequency: PWM频率 0: 1KHz 1: 12KHz 2: 24KHz 3: 48KHz default: 2
PWM duty cycle	0x20 R/W	Duty cycle																Duty cycle: 占空比 value: 0~100 default: 0
Fan RPM	0x30 R	Speed-L	Speed-H															Speed: 风扇每分钟的转速 value: 0~(11500±10%)
Fan signal frequency	0x40 R	Frequency-L	Frequency-H															Frequency: 风扇输出引脚频率
Flash write back	0xF0W	Write back																write 1, write back
Firmware Version	0xF0 R															Version		Version: 软件版本号
I2C Address	0xF0 R/W																Address	Address: I2C设备地址 value: 0x08~0x77 default: 0x18

寄存器说明
(1) Fan control：配置风扇的工作状态，默认状态为打开。
(2) PWM frequency：配置 PWM 信号输出的频率，默认值为 24kHz。
(3) PWM duty cycle：配置 PWM 信号输出的占空比，用于调节风扇转速。
(4) Fan RPM：读取风扇转速（RPM，即每分钟旋转次数）。
公式：Fan RPM = 60 × (Fan signal frequency / Pulse count)。
注：
Fan RPM 表示风扇每分钟的转速。
Fan signal frequency 是风扇输出引脚的信号频率。
Pulse count 表示风扇每转一圈产生的脉冲数（此处每圈产生 2 个脉冲）。
(5) Fan signal frequency：读取风扇输出引脚的信号频率。
(6) Flash write back：向该寄存器写入 1，将配置信息保存到内部 Flash 中。
配置内容包括：
Fan control
PWM frequency
PWM duty cycle
(7) I2C Address：设置 I2C 地址，地址设置成功后会立即生效，并保存到内部 Flash 中，掉电后不丢失。
注意事项
写入 Flash 时需先擦除内部 Flash，操作耗时较长，约 20ms。
为延长 Flash 使用寿命，请避免频繁写入。
如果要写入的值与内部 Flash 当前存储的值相同，则不会执行实际的写操作，从而减少不必要的 Flash 擦写，进一步延长其使用寿命。

Video

- Module Fan v1.1 Product Introduction and Case Demonstration

M013_V11_Module_Fan_v11_video.mp4

Product Comparison

Product Comparison		
	Module Fan v1.1	**Module Fan**
Communication	I2C (0x18)	/
Speed	11500RPM ±10%	7500RPM ±10%
Speed Control	Supported (via I2C protocol)	Not Supported